

Osprey transmission control malfunction P0700 from shift solenoid pack

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Area: bulletins

1 Condition

Affected Osprey vehicles may set diagnostic trouble code P0700 (Transmission Control System Malfunction) with an illuminated malfunction indicator lamp. Operators typically report harsh or delayed gear engagement, intermittent shift flare, and the transmission defaulting to a fail-safe limp-home mode. Before proceeding, confirm the complaint and review the Osprey Transmission Service Manual :: Diagnostic Trouble Code Reference to verify that P0700 is the active code rather than a stored history record. A general overview of the affected subsystems is available in the Osprey Transmission Service Manual :: Transmission System Overview .

2 Cause

The root cause has been traced to internal electrical degradation within the shift solenoid pack, part number TSOL-9520, where a failed solenoid winding interrupts the control signal monitored by the transmission control module. As the TSOL-9520 winding resistance drifts out of specification, the module reports the P0700 fault and commands the fail-safe strategy. This condition is distinct from torque converter clutch faults, which should be ruled out per the Osprey Transmission Service Manual :: Torque Converter Clutch Diagnosis . Follow the structured fault-tree in the Osprey Transmission Service Manual :: Transmission Fault Diagnosis Procedure before condemning the pack.

3 Repair Procedure

Replace the shift solenoid pack TSOL-9520 following procedure PROC-SHIFT-SOL, which covers fluid evacuation, valve body access, and electrical connector handling. Torque all fasteners to specification and clear the P0700 code, then road test to confirm normal shift quality. The complete step-by-step replacement is documented in the Osprey Transmission Service Manual :: Shift Solenoid Pack Replacement Procedure , with supporting subsystem detail in the Osprey Transmission Service Manual :: Shift Control Sub-System section. For broader diagnostic context refer to the Osprey Transmission Service Manual :: Diagnosis chapter.